

Sample Paper 7 — Class 5 Mathematics Olympiad (Level-2 Difficulty Step-Up)

Pattern: Same 50 questions / 60 marks / 60 minutes / no negative marking as Papers 1–6, but noticeably harder throughout — closer to SOF IMO **Level 2**. Attempt this only after scoring comfortably on Sample Paper 6.

Section 1 — Logical Reasoning (Q1–Q15, 1 mark each)

- Q1.** Find the odd one out: A) Mango B) Banana C) Carrot D) Apple
- Q2.** Author is to Book as Composer is to _____. A) Music B) Song C) Instrument D) Concert
- Q3.** If each letter of a word is moved 3 steps forward in the alphabet (wrapping Z → C), PEN becomes SHQ. Using the same rule, what does MATHS become? A) PDWKV B) PDWKU C) PDWLV D) QDWKV
- Q4.** This series follows two interleaved patterns — the 1st, 3rd, 5th, 7th terms, and the 2nd, 4th, 6th, 8th terms. Find the missing (8th) term: 3, 80, 7, 75, 11, 70, 15, ____ A) 63 B) 64 C) 65 D) 66
- Q5.** Five friends — Kabir, Leena, Madhav, Nisha, Omkar — sit in a row. Madhav sits exactly in the middle. Omkar sits at the left end. Nisha sits immediately to the right of Madhav. Leena sits between Omkar and Madhav. Find the seating order, left to right. A) Omkar, Leena, Madhav, Nisha, Kabir B) Omkar, Madhav, Leena, Nisha, Kabir C) Leena, Omkar, Madhav, Nisha, Kabir D) Omkar, Leena, Nisha, Madhav, Kabir
- Q6.** Tara walks 11 m East, then 8 m North, then 11 m West, then 8 m South. Where is she now relative to her starting point? A) Back at the starting point B) 8 m North C) 11 m East D) 19 m away
- Q7.** If the 4th day of a month is a Friday, what day of the week is the 29th of the same month? A) Monday B) Tuesday C) Wednesday D) Thursday
- Q8.** Find the odd one out: A) 8 B) 27 C) 64 D) 90 (Hint: three of these follow the same multiplication pattern — a number multiplied by itself twice.)
- Q9.** Find the next number: 3, 6, 18, 72, ____ A) 288 B) 324 C) 360 D) 432
- Q10.** In a row of students, Neha is 9th from the left and 14th from the right. If 3 students leave from the left end, what will be her new position from the left? A) 5th B) 6th C) 7th D) 8th
- Q11.** Find the odd one out: A) 17 B) 31 C) 42 D) 53
- Q12.** In a certain code, WATER is written as YCVGT (each letter moved 2 steps forward). How is GARDEN written in the same code? A) ICTFGP B) ICTGFP C) ICSFGP D) IDTFGP

- Q13.** Find the odd one out: A) Square B) Rhombus C) Trapezium D) Sphere
- Q14.** In a race, Rohan finished before Suresh but after Tarun. Vikas finished before Tarun. Who finished first? A) Rohan B) Suresh C) Tarun D) Vikas
- Q15.** Study the pattern: 2, 3, 5, 8, 12, _____. Find the next number. A) 16 B) 17 C) 18 D) 19

Section 2 — Mathematical Reasoning (Q16–Q35, 1 mark each)

- Q16.** $5,67,890 + 3,45,678 = ?$ A) 9,13,568 B) 9,13,468 C) 9,12,568 D) 9,14,568
- Q17.** $9,50,000 - 5,67,890 = ?$ A) 3,82,110 B) 3,81,110 C) 3,82,210 D) 3,83,110
- Q18.** $7,654 \times 12 = ?$ A) 91,848 B) 91,748 C) 91,948 D) 92,848
- Q19.** $1,09,608 \div 24 = ?$ A) 4,567 B) 4,557 C) 4,667 D) 4,467
- Q20.** Find the HCF of 84 and 126. A) 14 B) 21 C) 42 D) 63
- Q21.** Find the LCM of 18 and 24. A) 36 B) 48 C) 72 D) 144
- Q22.** Which of these fractions is the greatest: $\frac{6}{7}$, $\frac{9}{11}$, $\frac{5}{6}$, $\frac{13}{15}$? A) $\frac{6}{7}$ B) $\frac{9}{11}$ C) $\frac{5}{6}$ D) $\frac{13}{15}$
- Q23.** $3 \text{ km } 850 \text{ m} + 2 \text{ km } 950 \text{ m} = ?$ A) 5 km 800 m B) 6 km 700 m C) 6 km 800 m D) 6 km 900 m
- Q24.** A rectangular field is 42 m long and 26 m wide. Find its area. A) 1,082 sq m B) 1,092 sq m C) 1,102 sq m D) 1,112 sq m
- Q25.** A regular decagon (10 sides) has a perimeter of 120 cm. Find the length of each side. A) 10 cm B) 11 cm C) 12 cm D) 13 cm
- Q26.** The cost price of an item is ₹1,500 and the selling price is ₹1,950. Find the profit percentage. A) 25% B) 28% C) 30% D) 32%
- Q27.** A sum of money doubles itself in 5 years at simple interest. Find the rate of interest per annum. A) 15% B) 18% C) 20% D) 22%
- Q28.** ₹480 is divided among three people in the ratio 4:5:7. Find the largest share. A) ₹180 B) ₹190 C) ₹200 D) ₹210
- Q29.** Convert 0.375 to a fraction in simplest form. A) $\frac{3}{8}$ B) $\frac{3}{7}$ C) $\frac{2}{5}$ D) $\frac{1}{4}$
- Q30.** $5 \text{ kg } 250 \text{ g} + 3 \text{ kg } 800 \text{ g} + 2 \text{ kg } 950 \text{ g} = ?$ A) 11 kg 500 g B) 11 kg 900 g C) 12 kg D) 12 kg 100 g
- Q31.** How many seconds are there in 3.5 hours? A) 11,600 B) 12,000 C) 12,600 D) 13,000
- Q32.** Find the perimeter of an equilateral triangle with side 31 cm. A) 62 cm B) 93 cm C) 31 cm D) 124 cm
- Q33.** Find the area of a rectangle with length 42 cm and breadth 27 cm. A) 1,124 sq cm B) 1,134 sq cm C) 1,144 sq cm D) 1,154 sq cm

Q34. How many edges does a rectangular box (cuboid) have? A) 6 B) 8 C) 10 D) 12

Q35. Find the number of diagonals in a heptagon (7 sides). A) 12 B) 13 C) 14 D) 15

Section 3 — Everyday Mathematics (Q36–Q45, 1 mark each)

Q36. A school collected ₹2,25,000 from 9 classes equally. How much did each class contribute? A) ₹24,000 B) ₹24,500 C) ₹25,000 D) ₹25,500

Q37. A factory makes 2,800 bottles in 8 hours, at a steady rate. How many bottles does it make in 6 hours? A) 2,000 B) 2,050 C) 2,100 D) 2,150

Q38. A tank holds 4,500 L and is filled by a pipe at 180 L per minute. How long will it take to fill the tank completely? A) 23 min B) 24 min C) 25 min D) 27 min

Q39. A bus leaves city A at 6:55 a.m., stops for 22 minutes on the way, and reaches city B at 11:40 a.m. How much time did the bus spend actually travelling (excluding the stop)? A) 4 h 3 min B) 4 h 13 min C) 4 h 23 min D) 4 h 33 min

Q40. A chemist mixes 4 L 800 mL of one liquid with 2 L 200 mL of another, then pours the mixture equally into 5 bottles. How much does each bottle get? A) 1 L 200 mL B) 1 L 300 mL C) 1 L 400 mL D) 1 L 500 mL

Q41. Rohit buys 9 notebooks at ₹35 each and gets a discount of ₹31 on the total bill. How much does he pay? A) ₹274 B) ₹279 C) ₹284 D) ₹289

Q42. A cistern with 1,600 L of water is used for irrigation at the rate of $\frac{1}{10}$ of its original capacity every day. How much water remains after 3 days of use? A) 1,080 L B) 1,100 L C) 1,120 L D) 1,140 L

Q43. A delivery van carries 4 boxes weighing 19 kg 500 g, 17 kg 750 g, 14 kg 250 g, and 11 kg 500 g. Find the total weight carried. A) 62 kg B) 62 kg 500 g C) 63 kg D) 63 kg 500 g

Q44. 1,680 chairs are arranged equally into rows of 35 chairs each. How many rows will there be? A) 44 B) 46 C) 48 D) 50

Q45. A trader bought 90 kg of sugar for ₹3,600 and sold all of it at ₹46 per kg. Find his total profit. A) ₹500 B) ₹520 C) ₹540 D) ₹560

Section 4 — Achievers Section (Q46–Q50, 3 marks each)

Q46. A number is first multiplied by 7, then 28 is subtracted from the result, giving 126. Find the original number. A) 20 B) 21 C) 22 D) 23

Q47. The sum of three numbers is 200. The second number is twice the first, and the third is 20 more than the second. Find the first number. A) 32 B) 34 C) 36 D) 38

Q48. Anjali had a certain amount of money. She spent $\frac{3}{8}$ of it on books, then $\frac{1}{6}$ of what remained on stationery, and was left with ₹1,000. How much money did she have at first? A) ₹1,800 B) ₹1,860 C) ₹1,920 D) ₹1,980

Q49. In a school, the number of girls is 6 more than twice the number of boys. There are 318 students in total, and 104 of them are boys. Find the number of girls. A) 208 B) 210 C) 212 D) 214

Q50. A sequence starts with 5 and 8, and each later term is the sum of the two terms before it: 5, 8, 13, 21, 34, _____. Find the next term. A) 49 B) 52 C) 55 D) 58

Answer key: `sample-paper-07-answer-key.md`
